Gopal Gudipati
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EDUCATION

Gandhi Institute of Technology and Management — Vizag, India
Master of Technology in Computer Science and Engineering; GPA: 8.1 — *April 2024*

Dr. NTR College of Food Science and Technology — Bapatla, India
Bachelor of Technology in Food Technology; GPA: 7.0 — *August 2022*

SKILLS SUMMARY

- Languages: C++, Java, Python, Machine Learning
- Frameworks: Spring Boot
- Database: MySQL
- Tools: Jupyter Notebook, Visual Studio Code, GitHub
- Soft Skills: Excellent communication and Zeal to learn new skills.

PROJECTS

A Smart Air Pollution Detector Using Machine Learning — **Dec 23 - Feb 24**

- Built a Flask-based web application to detect air pollution levels in real time, integrating four supervised machine learning models like Logistic Regression, Random Forest, Decision Tree, and CatBoost to compare and optimize prediction accuracy.
- Achieved 92%+ model accuracy on test data by applying robust data preprocessing techniques, including outlier handling, normalization, and feature selection using correlation matrices.
- Deployed the application with an intuitive web interface, improving user engagement and enabling real-time input and feedback for pollution level detection. Reduced model latency by 40% through hyperparameter tuning and performance profiling, ensuring faster response times for predictions.
- Collaborated in a small team of 2–3 members, showcasing strong communication and full-stack development skills from backend ML integration to frontend UI design.

Smart Task Manager – Full Stack Web Application — **Jan 24 – Mar 24**

- Developed and deployed a full-stack task management application using React JS for the front end and Spring Boot with MySQL for the back end, supporting task creation, updates, and tracking by status, priority, and deadlines.
- Implemented secure RESTful APIs with Spring Boot and configured JWT authentication and role-based access control using Spring Security to manage user permissions and sessions.
- Designed responsive and interactive UI components with React JS, integrated Axios for API communication, and enabled real-time task data updates with dynamic filtering and sorting features.
- Performed unit and integration testing using JUnit and Mockito, achieving over 90% test coverage and enhanced backend performance by optimizing database queries with Hibernate/JPA.
- Utilized Git for version control, managed source code in GitHub, and configured CI/CD pipelines for continuous integration and deployment on Heroku/AWS, reducing manual deployment efforts.

CERTIFICATES

Programming in Python (Meta)

- Mastered fundamental Python syntax, proficiently utilizing control flow, loops, functions, and data structures.
- Acquired expertise in procedural programming paradigms and associated logical concepts, enhancing capabilities.

Introduction to Data Analyst (IBM)

- Learned about the data ecosystem, including the ETL process and big data basics.
- Mastered data gathering, identification, and cleaning for analysis preparation.

Foundations: Data, Data, Everywhere (Google)

- Created a comprehensive understanding of the data life cycle and various stages involved in the data analysis.
- Introduced diverse applications designed to streamline and optimize the data analysis journey, enhancing efficiency and accuracy.